

35. Sodium Chloride Measurements of water intake, obtained from a sample of 17 rats that had been injected with a sodium chloride solution, produced a mean and standard deviation of 31.0 and 6.2 cubic centimeters (cm^3), respectively. Suppose that the average water intake for noninjected rats observed over a comparable period of time is 22.0 cm^3 .

- Do the data indicate that injected rats drink more water than noninjected rats? Test at the 5% level of significance.
- Find a 90% confidence interval for the mean water intake for injected rats.

$$n = 17, df = 16, \bar{X} = 31, s = 6.2, \mu = 22$$

$$(a) H_0: \mu = 22, H_a: \mu > 22, \alpha = 0.05$$

$$t^* = \frac{31 - 22}{\frac{6.2}{\sqrt{17}}} \approx 5.985$$

$$t_{16, 0.05} = 1.746 \quad \text{reject: } t^* > 1.746$$

$$5.985 > 1.746 \Rightarrow \text{reject } H_0$$

$$(b) t_{16, 0.05} = 1.746$$

$$CI = \bar{X} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

$$= 31 \pm 1.746 \cdot \frac{6.2}{\sqrt{17}}$$

$$= 31 \pm 2.625$$

$$CI = (28.375, 33.625)$$